



DO-IT-YOURSELF: PROJECT

EFY Note

The source code of this project is available for free download at source.efymag.com

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the relay. The relay, in turn, rings the doorbell. The doorbell will, of course, not ring if the hand has not been sanitised.

For the doorbell to work, first the Arduino IDE needs to be installed

```

sketch_aug08a $
#define MQ3 A0
#define Thresold 420
int value;
void setup() {
  pinMode(MQ3, INPUT);
  pinMode(8, OUTPUT);
  pinMode(9, OUTPUT);
  Serial.begin(9600);
}
void loop () {
  value = analogRead(MQ3);
  Serial.println(value);
  if ( value > Thresold )
  {
    digitalWrite ( 8 , HIGH );
    digitalWrite ( 9 , HIGH );
  }
  else {
    digitalWrite(8, LOW);
    digitalWrite(9, LOW);
  }
  delay (500);
}

```

Fig. 4: The code snippet

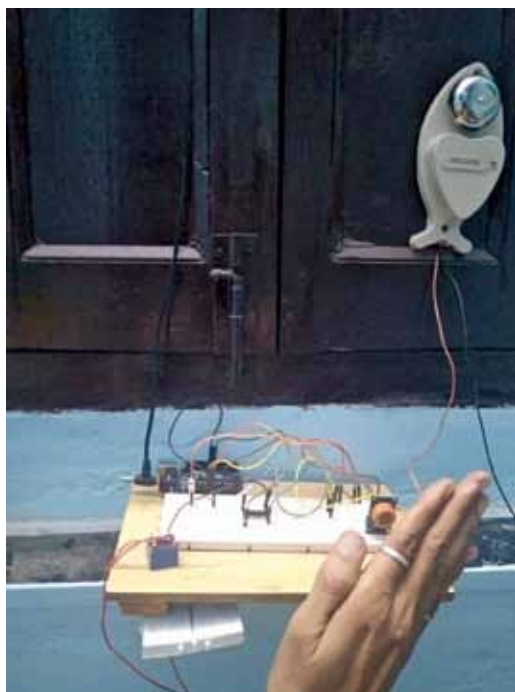


Fig. 5: Testing the circuit with sanitised hand

PARTS LIST

Semiconductors:

- Board1 - Arduino Board R3
- IC1 - ULN2003 relay driver
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistor (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm

Miscellaneous:

- CON1, CON2 - 2-pin connector
- CON3 - 4-pin connector
- BZ2 - Piezo-buzzer
- RL1 - 5V, 1C/O relay
- MQ3 alcohol sensor
- 5V regulated power supply

and then the code shown in Fig. 4 needs to be uploaded to the Arduino by selecting the right board and port number in Arduino IDE.

Construction and testing

The doorbell can be assembled by connecting all the components as per circuit diagram shown in Fig. 2. To test, first bring your unsanitised hand near the MQ3 sensor and see that the doorbell does not ring.

Next sanitise your hand and again bring it near the sensor. The doorbell should now ring (see Fig. 5). **EFY**

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